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NDVI-based split-window algorithm for precipitable water vapour retrieval from Landsat-8 TIRS data over land area

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An algorithm for the retrieval of precipitable water vapour (PWV) from Landsat-8 thermal infrared sensor (TIRS) data over land area has been developed in this paper. This method is based on the split-window covariance-variance ratio (SWCVR) theory and introduces normalized difference vegetation index (NDVI) to improve PWV retrieval from the relatively high-resolution thermal infrared data. Validation of the method is performed with meteorological data and Moderate Resolution Imaging Spectroradiometer (MODIS) total column PWV product (MOD05), and comparisons between NDVI-based SWCVR method and previous SWCVR method are employed. The root mean square error (RMSE) between PWV retrieved and that provided by meteorological data is 0.39 and 0.57 g cm^{-2} respectively for the proposed and previous method. The RMSE is 0.55 and 0.69 g cm^{-2} respectively for the proposed and previous method as validated with MOD05. It is concluded that the proposed method for retrieval of PWV from Landsat-8 TIRS data attained a better accuracy than the previous SWCVR method. PWV obtained from the proposed method is of great value as an input for land surface temperature retrieval and atmospheric correction for the Landsat-8 data.

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1. Introduction

Precipitable water vapour (PWV) is a key parameter to carry out processes such as atmospheric correction or retrieving atmospheric factors (García-Santos et al. 2012; Zhang, He, and Wang 2010). Knowledge of PWV is also necessary to improve the estimates of land surface temperature (LST) obtained from satellite data using split-window algorithms (Yu, Guo, and Wu 2014; Sobrino, El Kharraz, and Li 2003). Remote sensing is recognized as a useful way to acquire PWV at regional and global scales. Various algorithms have been proposed to retrieve PWV from satellite data. In general, these algorithms can be divided into three groups based on spectral region: (1) near-infrared techniques, (2) thermal infrared techniques, and (3) passive microwave techniques. The thermal infrared techniques are based on the ratio of split-window channel brightness temperature differences, which requires two nearby thermal infrared channels: one channel near to the water vapour absorption band and one nearby channel in the atmospheric windows (Kleespies and McMillin 1990; Jedlovec 1990). Based on split-window theory, Sobrino et al. (1994) proposed split-window covariance-variance ratio (SWCVR) technique. Li et al. (2003) refined the SWCVR technique by adding several constraints.

Landsat-8 was successfully launched on 11 February 2013; it continues the more than 40-year mission of acquiring global, moderate resolution images of Earth's surface every 16 days. Unlike prior Landsat satellites, the Landsat-8 satellite carries two instruments: (1)

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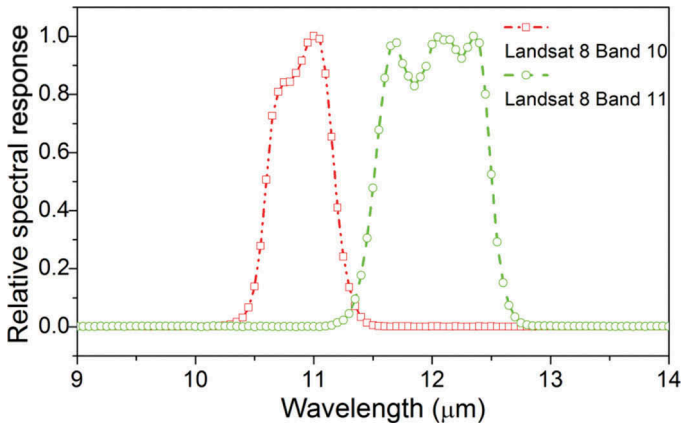

Figure 1. Landsat-8 TIRS bands' relative spectral response functions.

the operational land imager with nine spectral bands in the visible, near-infrared, and the shortwave infrared spectral region; and (2) the thermal infrared sensor (TIRS) with two thermal infrared spectral bands (band 10: 10.6–11.2 μm; band 11: 11.5–12.5 μm; see Figure 1). Two thermal infrared bands make contribution to retrieve PWV using SWCVR technique. However, the spatial resolution of 100 m for Landsat-8 TIRS data is obviously higher than previous thermal infrared data (ATSR, ATSR2) that SWCVR method has been applied to. The relatively high spatial resolution of TIRS data increases the spatial heterogeneity of land surface emissivity (LSE), which causes some loss of accuracy in the retrieval of PWV using the traditional SWCVR technique (analysis in Section 2.3). The aim of this paper is to introduce normalized difference vegetation index (NDVI) into the traditional SWCVR technique to account for spatial heterogeneity of LSE when using Landsat-8 TIRS data, hence improve the final PWV retrieval accuracy.

2. Methodology

2.1. Principle of the SWCVR method

Under the assumptions that atmosphere and LSE are unchanged, or effects of their spatial variations are not larger than the effect of instrument noise over the N neighbouring pixels, the SWCVR method (Li et al. 2003; Sobrino et al. 1994) is expressed as

$$w = a\left(\tau_j/\tau_i\right) + b. \tag{1}$$

$$\frac{\tau_j}{\tau_i} = \frac{\varepsilon_i}{\varepsilon_j} R_{j,i}, \text{ with } R_{j,i} = \frac{\sum_{k=1}^{N} (T_{i,k} - \bar{T}_i)(T_{j,k} - \bar{T}_j)}{\sum_{k=1}^{N} (T_{i,k} - \bar{T}_i)^2}. \tag{2}$$

$$r^2 = \frac{\left(\sum_{k=1}^{N} (T_{i,k} - \bar{T}_i)(T_{j,k} - \bar{T}_j)\right)^2}{\sum_{k=1}^{N} (T_{i,k} - \bar{T}_i)^2 \sum_{k=1}^{N} (T_{j,k} - \bar{T}_j)^2} = R_{i,j} R_{j,i}, \tag{3}$$

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where w represents PWV; a and b represent the coefficients of model of PWV; k represents pixel k; τ_i represents transmittance of TIRS band i; ε_i represents LSE of TIRS band i; $T_{i,k}$ represents brightness temperature of TIRS band i for pixel k; $\bar{T}_i$ represents mean brightness temperature of TIRS band i over N pixels; r denotes the linear correlation coefficient of two measurements T_i and T_j over the N neighbouring pixels, and is used to check whether the assumptions made in the derivation process of transmittance ratio are valid. Here, i and j represent respectively 10 and 11 for Landsat-8 data.

2.2. *The existing Landsat-8 PWV retrieval method in the literature (Ren et al.'s method)*

Using the traditional SWCVR method, Ren et al. (2015) proposed a PWV retrieval method for Landsat-8 TIRS data. Ren et al.'s method simply adapted the traditional SWCVR technique to the Landsat-8 TIRS data. However, the SWCVR technique is developed based on coarse spatial resolution satellite data (e.g. ATSR with 1.1 km spatial resolution), when used for relatively high spatial resolution data (e.g. Landsat-8 TIRS with a spatial resolution of 100 m, significantly higher than ATSR), it will inevitably have some limitations and improvements should be made.

2.3. *NDVI-based SWCVR method*

An important assumption that LSE are unchanged over the N neighbouring pixels is involved in the SWCVR method, which is reasonable for low-resolution satellite imagery, such as AVHRR with 1 km resolution, ATSR with 1.1 km resolution (Sobrino et al. 1996; Ottle et al. 1997). However, it is not very reasonable for relatively high-resolution satellite imagery, such as Landsat-8 TIRS.

The relatively high spatial resolution of TIRS data increases the spatial heterogeneity of LSE, which has a significant impact on determination of $R_{j,i}$. Under the condition that the atmosphere is unchanged over the neighbouring pixels, but where the LST and LSE change, the effects of LSE and LST differences on the $R_{j,i}$ were analysed by Sobrino et al. (1994). Figure 2 is constructed for several situations in which a pure soil pixel is adjacent to a mixed pixel or pure vegetation pixel. For Figures 2(*a*)–(*d*), the adjacent pixel is mixed with vegetation fractions of 0.25, 0.5, 0.75 and 1, respectively; therefore from Figures 2 (*a*) to (*d*), the LSE difference between the pure soil pixel and its adjacent pixel gradually increases. With the same PWV and LST difference (ΔT_s), error in $R_{j,i}$ clearly increases with increasing of the LSE difference (see Figures 2(*a*)–(*d*)); and it is shown that error in $R_{j,i}$ is less than 3% when a small difference of LSE is considered (Figure 2(*a*)); while error in $R_{j,i}$ can reach 12% when a significant difference of LSE is considered (Figure 2(*d*)).

The error in $R_{j,i}$ has a direct influence on the precision of PWV retrieved (Equations (1) and (2)). In order to obtain a higher accuracy PWV, error in $R_{j,i}$ must be reduced. It is found that there is a very high correlation between NDVI and LSE in the band covering 8–14 μm (Li et al. 2013). Therefore, NDVI is introduced to reduce the effect of LSE difference in the N neighbouring pixels, and hence reducing the error in $R_{j,i}$. In this way, the N neighbouring pixels is categorized into multiple groups using NDVI, which renders pixel-wise LSE of each group closer. The PWV for each group are respectively calculated; then PWV of the N neighbouring pixels are calculated using weighted sum method. In the following, a template extracted from the Landsat-8 image is used to represent the N neighbouring pixels.

The size of template is fixed to be $m \times m$ pixels. Atmospheric conditions are considered spatially unchanged at this size of template. The number of groups is set to n. Based on the

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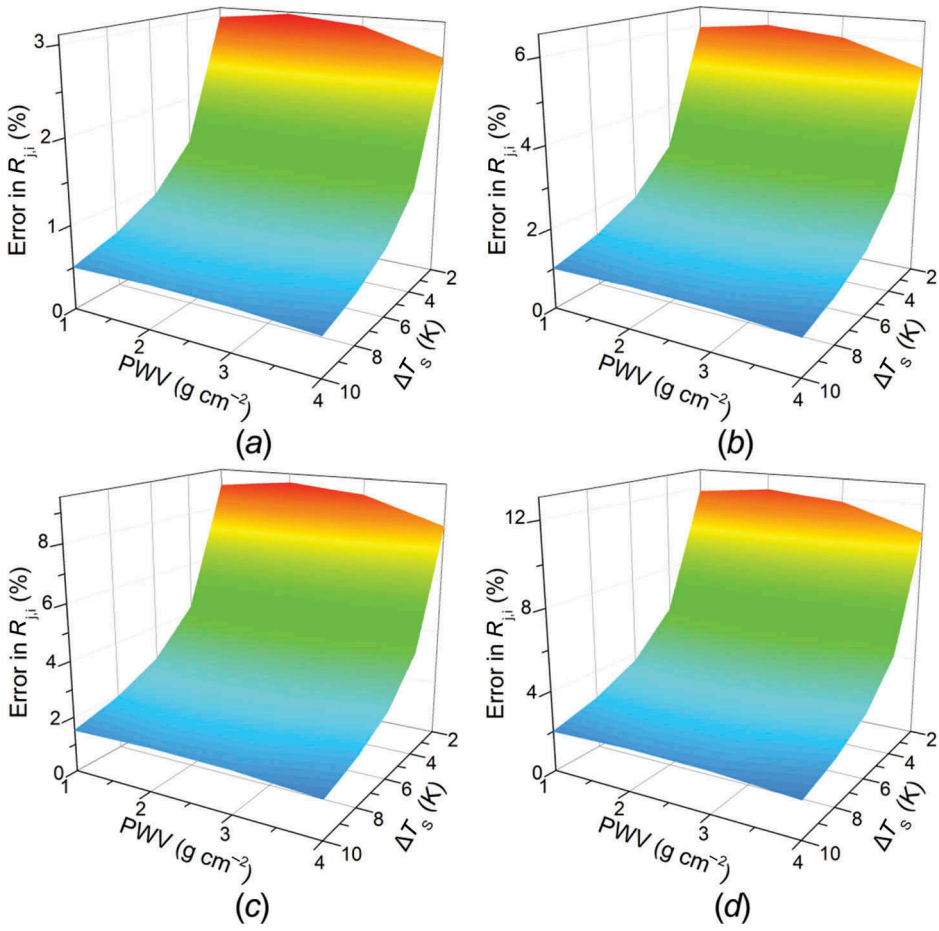

Figure 2. Error in $R_{j,i}$ considering PWV, differences of land surface emissivity (LSE) and land surface temperature (LST). The LSE difference gradually increases from (*a*) to (*d*).

SWCVR theory, the following procedures to retrieve PWV from Landsat-8 TIRS with the proposed method for given template are shown:

(1) Calculating brightness temperatures of TIRS channels 10 and 11 ($T_{10,k}$ and $T_{11,k}$) and NDVI (NDVI_k) for each pixel, k.
(2) Grouping pixels of the template based on NDVI_k and the number of groups. The NDVI interval of grouping ($\mathrm{NDVI_D}$) is calculated by dividing the difference between maximum NDVI ($\mathrm{NDVI_{max}}$) and minimum NDVI ($\mathrm{NDVI_{min}}$) of the whole image by the number of groups. Then, the value of dividing ($\mathrm{NDVI}_k - \mathrm{NDVI_{min}}$) by $\mathrm{NDVI_D}$ determines which group pixel k should belong to. i represents group number. Steps 3–5 are respectively employed for each group.
(3) Calculating mean brightness temperatures of TIRS channels 10 and 11 ($\bar{T}_{10}, \bar{T}_{11}$).
(4) Eliminating abnormal pixels by two restrictive conditions based on Li et al.'s (2003) study. First, the absolute value of (T_{10}–$\bar{T}_{10}$) must be greater than the absolute value of (T_{11}–$\bar{T}_{11}$); in addition, the value of (T_{10}–$\bar{T}_{10}$) and the value of (T_{11}–$\bar{T}_{11}$) should have the same sign. If one pixel meets these two

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restrictive conditions, then it is selected for calculations of $R_{10,11}$, $R_{11,10}$, r^2 and mean NDVI ($NDVI_m$), and $N_i = N_i + 1$ (N_i is the number of valid pixels that are used for computations). Conversely, pixels that do not meet these restrictive conditions are eliminated from the calculations.

(5) Calculating values of $R_{10,11}$, $R_{11,10}$ and r^2 using Equations (2) and (3). LSE ratio ($\varepsilon_{10}/\varepsilon_{11}$) is determined from $NDVI_m$ with Equation (4). Transmittance ratio (τ_{11}/τ_{10}) is calculated by multiplying LSE ratio ($\varepsilon_{10}/\varepsilon_{11}$) by $R_{11,10}$. The PWV for the group i (w_i) is determined from transmittance ratio using Equation (1). If value of r^2 is greater than 0.97, w_i is valid; otherwise w_i is invalid, and make N_i equal to zero.

(6) PWV (w_i) and the number of valid pixels (N_i) for each group are acquired with steps 3–5. The comprehensive PWV (w) for the whole template is calculated with the weighted sum method based on w_i and N_i (Figure 3).

2.3.1. *Relationship between the transmittance ratio and PWV*

The relationship between transmittance ratio and PWV can be derived from simulated data. Atmospheric transmittance can be simulated by the atmospheric radiative transfer models, such as moderate resolution atmospheric transmission (MODTRAN) (Berk et al. 1999). In this study, MODTRAN 4.0 is used to simulate the atmospheric transmittances of channels 10 and 11 with relative spectral response of the TIRS instrument and TIGR3 atmospheric profiles as input. The Thermodynamic Initial Guess Retrieval (in its latest version, TIGR3) data set is a climatological library of 2311 representative atmospheric situations, including 872 tropical atmospheres, 742 mid-latitude atmospheres and 697 polar atmospheres. Each atmosphere is described by the values of the atmospheric temperature, water vapour and ozone concentration on a given pressure grid from the surface to the top of the atmosphere. The PWV in TIGR3 varies from 0.066 to 7.833 g cm^{-2}. Figure 4(*a*) represents the histograms of PWV. The worldwide set of atmospheric situations (TIGR3) is credible to derive the global relationship between the transmittance ratio and PWV (Li et al. 2003). Thus these 2311 atmospheres are used to create the simulated data; then the transmittance ratio of Landsat-8 TIRS bands 11 and 10 (τ_{11}/τ_{10}) is regressed against PWV.

As Figure 4(*b*) shows, there is a strong correlation between transmittance ratio and PWV except for an inflection point around 0.9. In order to obtain better fitted model between PWV and transmittance ratio, transmittance ratio was divided into two ranges: less than 0.9 and larger than 0.9. Then the relationships between transmittance ratio and PWV (w) in each range were regressed respectively. Results are shown in Table 1.

2.3.2. *Determination of the LSE ratio*

In the SWCVR method, the LSE ratio of two TIR bands is assumed to be a constant of 1. In this proposed method, the NDVI threshold method developed by Sobrino et al. (2008) is employed to obtain LSE of TIRS bands 10 and 11; then the LSE ratio is calculated. Vegetation and soil LSE for Landsat-8 TIRS bands were calculated from the Moderate Resolution Imaging Spectroradiometer (MODIS) University of California, Santa Barbara, emissivity library. Then the LSE ratio of TIRS channels 10 and 11 was finally given:

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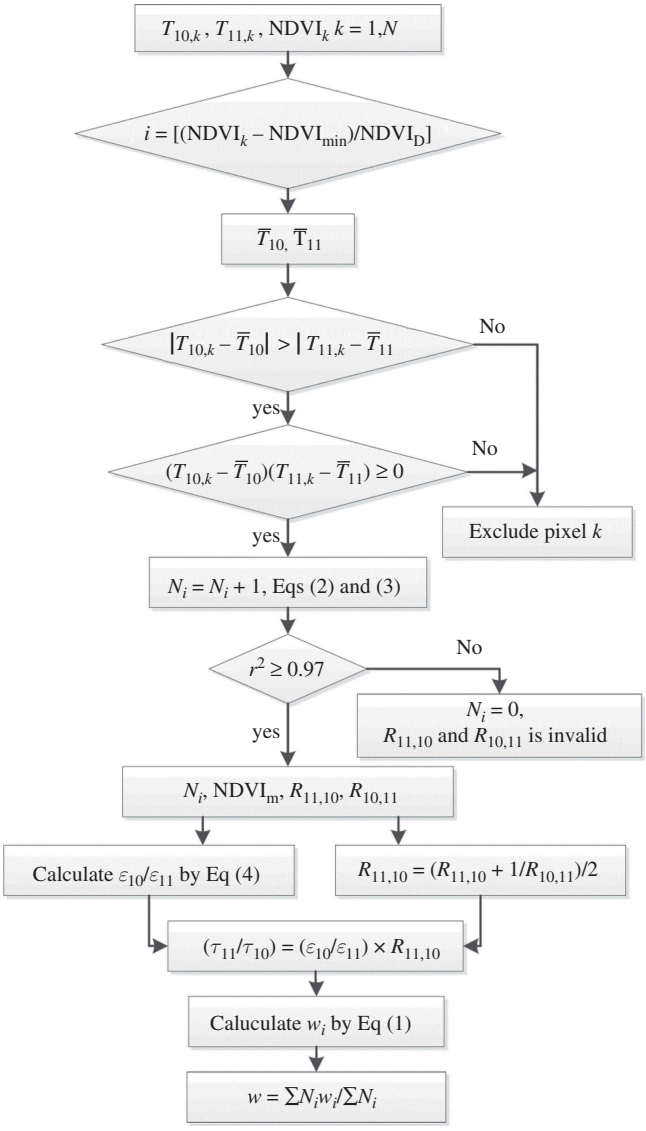

Figure 3. Flow chart of the procedure to retrieve precipitable water vapour from Landsat-8 TIRS data for a given template.

$$\frac{\varepsilon_{10}}{\varepsilon_{11}} = \begin{cases} 0.9939 & \mathrm{NDVI} < \mathrm{NDVI_S} \\ (0.0195P_\mathrm{V} + 0.9688)/(0.0149P_\mathrm{V} + 0.9747), & \mathrm{NDVI_S} < \mathrm{NDVI} < \mathrm{NDVI_V} \\ 0.9966, & \mathrm{NDVI} > \mathrm{NDVI_V}. \end{cases} \tag{4}$$

$$\text{with } P_\mathrm{V} = \left(\frac{\mathrm{NDVI} - \mathrm{NDVI_S}}{\mathrm{NDVI_V} - \mathrm{NDVI_S}}\right)^2,$$

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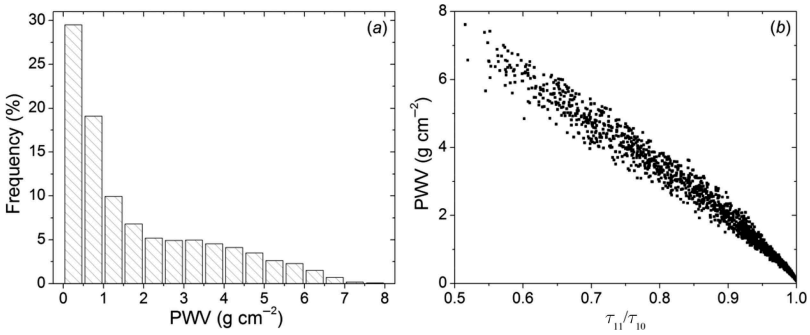

Figure 4. (*a*) Histograms of precipitable water vapour (PWV) of 2311 TIGR atmosphere profiles; (*b*) the relationship between PWV and the transmittance ratio of 2311 TIGR atmospheric profiles.

Table 1. Relationship between precipitable water vapour and the transmittance ratio for Landsat-8 TIRS data.

τ_{11}/τ_{10}	Expression for w	R^2
Larger than 0.9	$w = -18.973(\tau_{11}/\tau_{10}) + 19.13$	0.9663
Less than 0.9	$w = -13.412(\tau_{11}/\tau_{10}) + 14.158$	0.9366

where $NDVI_S$ corresponds to NDVI of bare soil, $NDVI_V$ corresponds to NDVI of the fully vegetated area and P_V represents vegetation fraction. $NDVI_V$ and $NDVI_S$ values can be extracted from the NDVI histogram.

3. Results and discussion

In this study, 10 Landsat-8 images of two places were used to determine PWV using the method proposed by Ren et al. (2015) and the NDVI-based SWCVR method proposed in this study. One place is in Beijing, east of China (named P1, path/row: 123/32; 39.3°N–41.3°N and 115.3°E–118.1°E), the other is in Shanxi province, west of China (named P2, path/row: 127/36; 33.5°N–36.0°N and 107.5°E–110.0°E). The acquisition time of Landsat-8 images are distributed in different seasons. The template was fixed to 20 × 20 pixels for both Ren et al.'s method and the proposed method; the number of groups was set to 3 for this proposed method. Meteorological observations from 19 weather stations were used to perform validation to the PWV retrieval results. In addition, coincident MODIS MOD05 PWV product was also employed to perform per-pixel comparisons with both PWV retrieval results. In the following, PWV calculated using the method proposed by Ren et al. is termed as PWV_R, PWV calculated using the NDVI-based method proposed in this paper is termed as PWV_N, PWV calculated from MOD05 data is termed as PWV_{MO} and PWV calculated from meteorological data is termed as PWV_{ME}.

Table 2 represents the root mean square error (RMSE) between PWV_R, PWV_N and PWV_{ME}. As can be seen from Table 2, the RMSE is between 0.22 and 1.03 g cm^{-2} for PWV_R, and is between 0.11 and 0.72 g cm^{-2} for PWV_N. RMSE between PWV_R and PWV_{ME} is larger than that between PWV_N and PWV_{ME} in most cases, and the average RMSE is 0.57 and 0.39 g cm^{-2} for Ren et al.'s method and the proposed method,

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Table 2. The RMSE between PWV_R and PWV_{ME} and between PWV_N and PWV_{ME}.

Place	Date	RMSE between PWV_R and PWV_{ME} (g cm^{-2})	RMSE between PWV_N and PWV_{ME} (g cm^{-2})
P1	12 May 2013	0.31	0.42
	29 April 2014	0.22	0.11
	15 May 2014	0.42	0.23
	4 September 2014	0.63	0.43
	6 October 2014	0.31	0.23
P2	30 July 2014	0.55	0.31
	2 October 2014	0.55	0.39
	3 November 2014	0.98	0.72
	2 December 2014	1.03	0.71
	28 April 2015	0.67	0.30
Average		0.57	0.39

respectively. Validation results with meteorological data show the NDVI-based method achieved an overall better accuracy than Ren et al.'s method.

Furthermore, MOD05 data were used to perform cross-comparison. Comparisons between PWV_R, PWV_N and PWV_{MO} were carried out pixel by pixel. Figure 5 represents the RMSE and coefficients of determination (R^2) between PWV_R, PWV_N and PWV_{MO}. As Figure 5 shows, the RMSE between PWV_N and PWV_{MO} are obviously lower than that between PWV_R and PWV_{MO}; and the R^2 between PWV_N and PWV_{MO} are higher than that between PWV_R and PWV_{MO}. It is also indicated that the proposed method attained a better accuracy than Ren et al.'s method.

Both the validation results from meteorological data and MODIS PWV products show that the method proposed in this paper performed better than the previous method through considering the spatial heterogeneity of LSE. Grouping pixels of the template based on NDVI is the main difference between Ren et al.'s method and this proposed method, which can reduce the effect of the spatial heterogeneity of LSE to some extent. As can be seen from Table 1, an error in transmittance ratio leads to 13.4–19.0 times error in PWV. Transmittance ratio of two TIR bands is calculated by multiplying $R_{11,10}$ by LSE ratio of two TIR bands, thus the calculation accuracy of transmittance ratio depends on the accuracy of $R_{11,10}$. Therefore, it can also be said that accuracy of PWV is dependent on accuracy of $R_{11,10}$. Error in $R_{11,10}$ increases with increasing of LSE difference (see Figure 2), indicating that accuracy of PWV can be improved by reducing LSE difference, which is confirmed by the validation results as presented in Table 2 and Figure 5.

It should be noted that an important advantage of the proposed method is that it can provide PWV from Landsat-8 image itself; then the PWV can be used as an input parameter for LST or land surface reflectance (LSR) retrieval from the same Landsat-8 data. With the proposed PWV retrieval method, PWV can be obtained and inputted to the Landsat-8 LST retrieval algorithm, thus LST can be derived from the Landsat-8 data itself requiring no external data.

4. Conclusion

In this study, an NDVI-based split-window algorithm for PWV retrieval from Landsat-8 TIRS data has been proposed. Using meteorological data and the MODIS PWV product,

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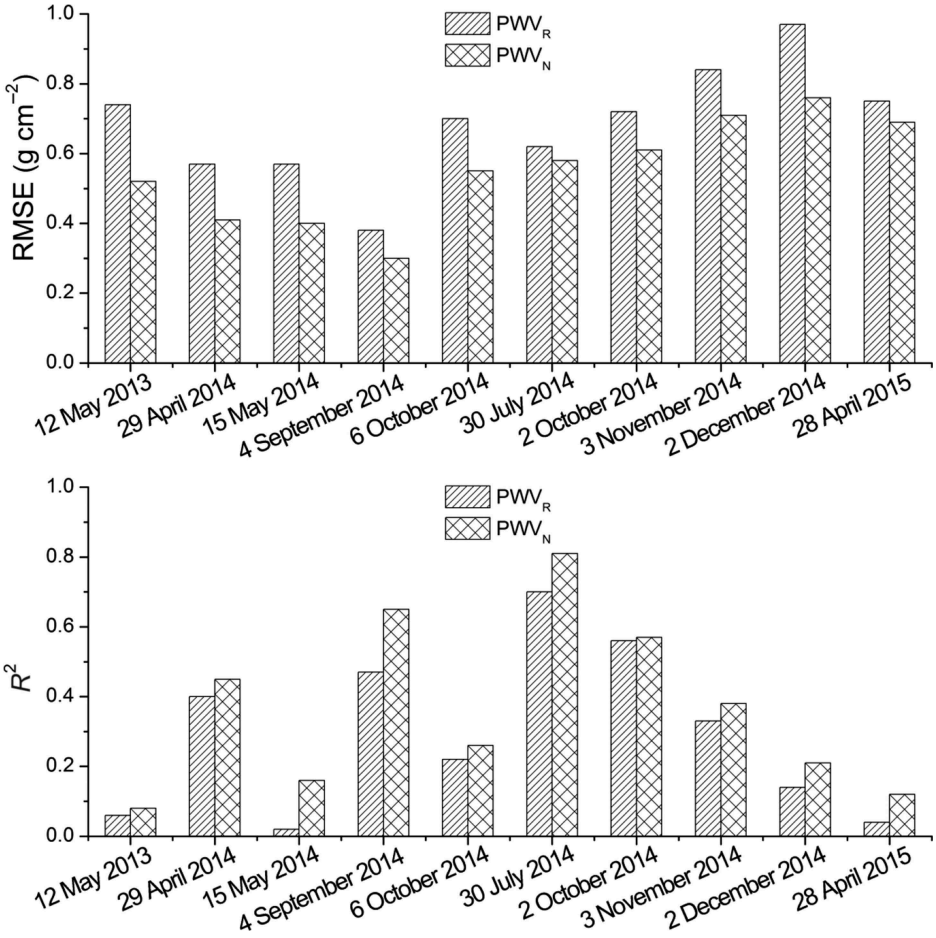

Figure 5. RMSE and R^2 between PWV_R and PWV_{MO} and between PWV_N and PWV_{MO} for different Landsat-8 images; the first five dates correspond to P1, and the last five dates correspond to P2.

performance of the proposed method was assessed and compared. Based on the validation results of this study, it is concluded that by introducing NDVI into SWCVR the proposed method can provide better estimates of PWV from relatively high-resolution thermal infrared data (Landsat-8 TIRS data). PWV obtained from the proposed method is important for LST and LSR retrievals from the Landsat-8 data.

Disclosure statement

No potential conflict of interest was reported by the authors.

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product (MOD05) were downloaded from the MODIS atmosphere website at http://modis-atmos.gsfc.nasa.gov/.

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